Statistical Foundations for Reliable AI Reasoning under Compute Constraints

Brief topic description for Program Officer discussion — ONR FY27 YIP (N0001426SF004)

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Problem. At inference time, language models can sample reasoning paths, score intermediate steps with process reward models (PRMs), and retrieve more evidence. Yet common rules such as majority vote and fixed best-of-N implicitly give extra weight to repeated reasoning, even when samples share an error. Consider a shipboard fuel-range estimate: fifteen of twenty generated solutions reuse a misread consumption rate from the same log entry. The apparent consensus grows, but the underlying evidence does not. How much independent information does another reasoning trace actually provide, and when is an independent check worth more than another sample?

Central hypothesis. We hypothesize that explicitly modeling shared errors among reasoning traces and their verifiers will improve accuracy at a fixed computation budget and support better-calibrated abstention. The statistical challenge is to determine what latent step correctness and error dependence can be identified from few labeled examples, and what risk-control guarantees remain valid under stated assumptions.

Research plan.

- **Estimate information in dependent traces.** Model the validity of each reasoning step as a latent variable and each verifier score as a noisy measurement. Shared prefixes, retrieved sources, and model-specific errors create dependencies among paths. Develop structured estimators of the information gained from an additional trace, characterize identifiability with limited labels and abundant unlabeled samples, and quantify uncertainty when dependence cannot be reliably estimated.
- **Allocate computation through sequential decisions.** Treat sampling another solution, verifying a specific step, retrieving independent evidence, and stopping or abstaining as actions with different costs. Develop value-of-information policies that trade expected error reduction against tokens, tool calls, and latency. Study selective-risk control across questions, making explicit the assumptions needed for finite-sample bounds and testing how those bounds fail under distribution shift.
- **Learn complementary process verifiers.** Extend synthetic-error supervision to factual, logical, and evidence-grounding checks, and measure when verifiers share rather than correct one another's mistakes. Combine their scores using the dependence model instead of ad hoc averaging. Identify regimes in which further sampling or checking adds cost but cannot resolve a common faulty premise.

Evaluation and expected outcomes. Test on MATH, MEQA, and MuSiQue, together with controlled cases where reasoning paths deliberately share prefixes or incorrect evidence. Compare against self-consistency, verifier-ranked best-of-N, and adaptive sampling heuristics at matched tokens and tool costs. Report accuracy–computation curves, error among accepted answers at matched coverage, calibration, and empirical validity of risk targets under dependent samples and domain shift.

Naval relevance. A shipboard planning assistant may appear confident about fuel range because many candidate calculations rely on the same rate copied from one report. Its best next step may be to verify that rate rather than generate more calculations. The research addresses Mathematical Data Science priorities in principled information integration, decision making under uncertainty, and fundamental trade-offs between inference accuracy and limited computing resources.

Prior work and PI qualifications. FG-PRM (Li et al., 2025, Findings of EMNLP) provides step-level error supervision; PRD (Li et al., 2024, TMLR) studies imperfect model judgments; and HiPRAG (Wu et al., 2026, ICLR) studies efficient agentic retrieval. The PI holds a 2024 NSF CAREER award and received a 2023 Amazon Research Award for process-guided fine-tuning. As a postdoc, the PI led system integration for the RESIN-11 team in DARPA KAIROS.